USER MANUAL

Smart Access Control Dashboard — Guide for Admins & Instructors

1. Who This Is For

This guide covers day-to-day use of the web dashboard by the two roles the system supports:

Role	Can do
Instructor	View door status, alerts, and access logs (read-only)
Admin	Everything an instructor can, plus: override door locks, resolve alerts, and manage users/credentials/schedules via the API

2. Logging In

1. Go to the dashboard URL your administrator gave you (e.g. http://<host>:5173 in development, or wherever it's deployed).
2. Enter your email and password and select Sign In.
3. After 5 incorrect attempts on the same account within a few minutes, the system temporarily locks that login out and tells you how long to wait — this is a deliberate anti-brute-force protection, not a bug. Wait it out or contact your admin.
4. Your session stays signed in for about an hour; after that you'll be asked to log in again.

3. The Main Dashboard

After logging in you land on the main dashboard, which refreshes itself automatically every 5 seconds — you don't need to reload the page to see current status.

3.1 Alert banner

Unresolved alerts (tamper, forced entry, door propped open, offline, access denied, authentication failure) appear at the top of the page. Admins see a Resolve button on each one; instructors see the same list read-only. Resolving an alert marks it handled — it doesn't undo whatever triggered it, so make sure the underlying issue (e.g. a door actually propped open) is actually addressed first.

3.2 Door status grid

Each door appears as a card showing:

- Online / offline — whether the door node (or the Building Gateway relaying it) has reported in recently. If a door has been offline longer than expected, check that node's power/network before assuming a dashboard glitch — see Section 5.
- Locked / unlocked — current state as last reported.
- An unlock / lock button (admins only) — see Section 4.
- A view logs link, which loads that door's access history into the table below (Section 3.3).

3.3 Access log table

Selecting a door loads its most recent access events (up to 200): timestamp, method (card, card+fingerprint, remote override, or REX/exit button), and result (granted or denied). This is the same data an auditor or investigator would want after an incident — export it via your browser's print-to-PDF or copy the underlying data via GET /api/doors/{id}/logs if you need it in another format (see API_Reference.docx).

4. Overriding a Door (Admins Only)

Use this to unlock a door remotely (e.g. a visitor locked out, an instructor without their card) or to force-lock one (e.g. responding to an incident).

5. Find the door's card in the status grid.
6. Select Unlock or Lock.
7. The dashboard shows whether the command was delivered to the door (mqtt_delivered). If it wasn't delivered, the broker or that door's connection is down — the attempt is still logged for audit purposes either way.

Note: An override is logged as its own access event (method: override) distinct from a card/fingerprint read, so the audit trail always shows whether a door was opened by a credential or by an admin's remote command.

5. Understanding Door Status Delays

A door's online/offline status is based on when it last actually reported in, not a live video feed — there is an inherent delay, typically a few seconds under normal conditions. If a door (or several doors that share one RS-485 gateway) suddenly show offline together, the most likely cause is the shared gateway or network link, not every individual door node failing at once — check the gateway/network before assuming multiple simultaneous hardware failures.

As of this system's Phase 7 testing, a door that stops reporting (including due to a crashed gateway) is automatically marked offline within about 30-40 seconds even if nothing explicitly says so — you should never see a door that's actually been silent for an hour still showing as "online."

6. Managing Users, Credentials & Schedules

The dashboard's web UI currently covers monitoring and override only (Sections 3-4). Administrative tasks that change who has access are done through the backend's interactive API page rather than a dedicated screen — this is a real, current limitation of the UI, not a missing feature of the system underneath it:

- Adding a user or issuing/revoking a card credential: use POST /api/users, POST /api/credentials, or DELETE /api/credentials/{id} via http://<backend-host>:8000/docs (log in there is the same admin token used by the dashboard).
- Setting up a door's access schedule (which hours/days it should unlock automatically for a class): POST/PUT /api/schedules, same interface.

Full request/response details for all of these are in API_Reference.docx. A future iteration of this project would be a good place to add dedicated dashboard screens for these — noted as a real next step, not glossed over.

7. Troubleshooting

Symptom	Likely cause / what to do
Login says "Too many failed attempts"	Rate limiting after 5 wrong passwords — wait for the stated cooldown, or confirm you have the right password with your admin
A door shows offline and won't come back	Check that door's power and network/RS-485 connection; if several doors on one gateway are all offline together, check the gateway host first
Unlock button says mqtt_delivered: false	The MQTT broker or that door's connection is currently down; the attempt is logged, but the door likely never received the command — try again once connectivity is confirmed
An alert won't go away after resolving it	Refresh — the dashboard polls every 5 seconds, so there can be a brief delay before a resolved alert disappears from the list
I need to add a new card or user and don't see how	That's done via the API docs page, not the dashboard yet — see Section 6